Terra-Alert Landslide Model: Status and Plan

Active model: v3 (XGBoost, NASA IMERG rain) | 13 November 2026

Summary

- The v3 model is **fair but weak**: on areas it never saw during training, it ranks a landslide day above a safe day 59% of the time (ROC-AUC 0.59; 0.5 = coin flip).
- The older v2 model showed **90% accuracy**, but that came from **data leakage**. Tested fairly, it scored ROC-AUC 0.49, worse than guessing.
- Four fixes are done: alert cut-offs chosen for **70% recall** (the old alert level caught 41%), a plains safety rule, fail-safe handling of missing data, and a check of live vs training rain data.
- **80%+ accuracy is not reachable honestly with the current data.** The realistic path is better landslide data plus a two-part model ("where" x "when"), described in section 6.

1. What the model does, and why it is hard

For a place and a day, the model reads 20 inputs (rain over the last 3/7/14/30 days, slope, elevation, vegetation, soil) and returns a risk score. It was trained on 1,610 examples from Himachal, Uttarakhand and Kashmir (2006-2017):

Example type	Rows	What it teaches
Landslide (label 1)	552	Location and date of a recorded landslide
Same place, other year	552	Same spot, same calendar day, a year without a landslide: *what was different that day?*
Nearby, same day	506	5-30 km away, similar height, same day: *why did it slide here and not there?*

The hard part: rain data comes in squares about **10 km wide**, and catalogue locations can be off by a few km. A slope that slid and the slope next to it often look identical to the model. Median values:

	Elevation	Rain, last 3 days	Rain, last 30 days
Landslide	1,531 m	31 mm	255 mm
Same place, other year	1,531 m	7 mm	183 mm
Nearby, same day	1,533 m	32 mm	260 mm

Rain clearly separates a landslide day from a normal day at the same place (ROC-AUC 0.635). Nothing separates neighbouring slopes on the same day (ROC-AUC 0.54).

2. Why "accuracy" is the wrong target

- **Class imbalance:** 2 of every 3 rows are safe. A model that *never* warns scores 67% accuracy while catching zero landslides.
- **v2 is the warning example.** Its "safe" points included plains and glaciers (228 of 555 below 800 m, some above 6,000 m) and had no dates. It learned "hills = danger", not rain or slope, which gave 90% accuracy on a random split but 0.49 ROC-AUC on unseen areas.
- For a warning app, a **missed landslide is far worse than a false alarm**, so we choose the cut-off by **recall** and report the false-alarm cost openly.

v3 at the old and new alert cut-off (measured on areas it never trained on):

	Old alert level (score >= 40)	New cut-off (MODERATE)
Landslides caught (recall)	41%	**70%**
Alerts that were real (precision)	41%	40%

	Old alert level (score >= 40)	New cut-off (MODERATE)
Safe places wrongly alerted	31%	55%
Accuracy	59%	53%

Accuracy dropped by 6 points, and in exchange the app catches **1.7x more landslides** (70% vs 41%). The model itself did not change: same ranking of places, same ROC-AUC 0.59. Only the alert cut-off moved.

3. Fixes completed

Fix 1: alert cut-offs chosen on purpose

v3 learned from 1 landslide per 2 safe rows, so its raw output is not a real-world probability. Cut-offs were chosen from **out-of-fold** scores (each row scored by a model that never saw its area) and mapped onto the app's existing 40 / 60 / 80 bands, so the rest of the app needed no changes.

Level starts at	Raw cut-off	Landslides caught	Alerts that were real	Safe rows flagged
MODERATE (alerts)	0.21	70%	40%	55%
HIGH	0.40	40%	41%	30%
CRITICAL	0.67	15%	46%	9%

Fix 2: plains safety rule turned back on

Only 27 of 1,610 training rows come from plains (Haryana, Delhi, UP), so v3 never learned flat land and could flag Delhi in the monsoon from rain alone. Flat, low areas (below 800 m, slope under 15 degrees) now always show LOW. Checked: Delhi with 150 mm of rain now shows 5 / LOW, and Shimla with the same rain shows 64 / HIGH.

Fix 3: live rain data checked against training rain data

- v3 was trained on IMERG V07 rain. Live predictions also use V07 (confirmed). The backup source is now V08, then Open-Meteo, and every prediction records which source it used.
- **Measured:** in recent months the Open-Meteo backup reported only **38-56%** of V07's rain. Swapping it in changed the risk level for **55%** of locations and flipped the alert for **29%**. It is not safe to use without a warning.
- **Fail-safe:** if rain or terrain data is missing, the app now shows UNKNOWN. Before, it silently used 0 mm of rain or 0 m elevation, which showed as "LOW risk".

Fix 4 (experiment, not yet in the app): small-scale terrain features

Seven new inputs (slope at the exact point, steepest slope nearby, local relief, ridge vs valley position, curvature, height above the nearest stream, upstream drainage area), tested 5 times with different splits:

	ROC-AUC, all rows	ROC-AUC vs nearby same day	False alarms at 70% recall
v3 inputs	0.577 +/- 0.007	0.533 +/- 0.007	59%
+ terrain features	0.598 +/- 0.008	0.566 +/- 0.007	57%

The gain is real (about 4-5 times the run-to-run spread) but small. Landslide spots sit lower above streams than nearby safe spots (63 m vs 103 m). Some of this may be reporting bias: roads follow valleys, and the catalogue mostly records landslides that hit roads.

4. Status against our goals

Goal	Status	What is left
Good recall	**Done**: 70% (was 41%)	Can be set to 80% at the cost of more false alarms

Goal	Status	What is left
No "hallucination" (confident nonsense)	**Mostly done**	Missing data now shows UNKNOWN; plains rule on. Next: flag inputs outside the training range
No bias affecting output	**Fair across hill states**	Catch rate 69-73% in Himachal, Kashmir and Uttarakhand. Remaining bias comes from the news-based landslide catalogue
No overfitting	**Not yet**	v3 memorises training rows (121 of 400 scored CRITICAL). Fix: simpler trees tuned with spatial cross-validation
High, honest scores	**Not yet**: ROC-AUC 0.59	Needs better data and the two-part model (section 6)

5. Known limits of the current data

- **Small and old:** about 550 landslides from 2007-2016, so recent disasters (Himachal 2023) are missing.
- **Location precision:** catalogue points can be a few km off, which blurs slope and terrain values.
- **Reporting bias:** the catalogue is built from news, so landslides in remote valleys are under-recorded.
- **Mostly a rain detector today:** the median 3-day rain is 31 mm when v3 alerts and 6 mm when it does not, so it warns on most rainy days in the hills.

6. Plan

- **Step 1: fix overfitting and broken inputs** (quick). Remove the input that is constant in every row (*north_facing*), cap the unstable rain-acceleration input, and tune tree depth and count inside spatial cross-validation. Expect a more trustworthy model, not a big score jump.
- **Step 2: better landslide data** (largest impact). The Geological Survey of India landslide inventory (Bhukosh portal) is much larger, more precise and more recent. It has to be downloaded by a team member.
- **Step 3: two-part model, built as v5.** A *susceptibility map* ("where": which slopes are dangerous, from terrain, rock type and distance to rivers) combined with a *rain trigger* ("when": enough recent rain to set them off). Final risk = dangerous slope x heavy rain. Each half is an easier, better-defined question than the one v3 tries to answer at once.
- **Reporting rule from now on:** every model is judged on **unseen areas** (spatial cross-validation) and reported with ROC-AUC, recall and false-alarm rate, never with random-split accuracy alone.

Decision needed: when live rain falls back to Open-Meteo, should the app show "rain data delayed" (no alerts), or show the score with a "low confidence" badge? Recommendation: the badge, plus a few weeks of logging to see how often the fallback actually happens.

Glossary

Term	Meaning
Recall	Of all real landslides, the share the model warned about. The most important number for a warning app.
Precision	Of all warnings raised, the share that were real landslides.
False-alarm rate	Of all safe places/days, the share that wrongly got a warning.
ROC-AUC	How well the model ranks risky above safe, ignoring the cut-off. 0.5 = coin flip, 1.0 = perfect.
Spatial cross-validation	Testing on whole geographic areas the model never saw in training, so it cannot pass by memorising places.
Out-of-fold scores	Scores where every row comes from a model that did not train on that row's area.

Term	Meaning
Overfitting	The model memorises its training rows instead of learning general patterns, so it looks great in training and fails on new data.
Data leakage	The test contains an easy shortcut (like v2's plains and glacier points), so the score looks high but means nothing.

Code and data: *ai-module/v3/* (calibrate_thresholds_v3.py, compare_rain_sources_v3.py) and *ai-module/v5/terrain_experiment_v5.py*; backend changes in *backend/services/*.